

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression, Ridge,
Lasso
from sklearn.metrics import r2_score
from scipy import stats
import warnings
warnings.filterwarnings('ignore')

df = pd.read_csv('/content/updated_pollution_dataset.csv')

print("Initial data shape:", df.shape)
print(df.head())

print("\nMissing values per column:")
print(df.isnull().sum())

df = df.drop_duplicates()
df = df.dropna()

numeric_cols = df.select_dtypes(include=[np.number]).columns
z_scores = np.abs(stats.zscore(df[numeric_cols]))
```

```
df = df[(z_scores < 3).all(axis=1)]
```

```
print(f"\nShape after removing outliers: {df.shape}")
```

```
print("\nDataset columns:")
```

```
print(df.columns)
```

```
print("\nDataset info:")
```

```
df.info()
```

```
plt.figure(figsize=(10, 5))
```

```
sns.heatmap(df.isnull(), yticklabels=False, cbar=False,  
cmap='viridis')
```

```
plt.title('Missing Data Heatmap')
```

```
plt.show()
```

```
corrmat = df[numeric_cols].corr()
```

```
plt.figure(figsize=(20, 20))
```

```
sns.heatmap(corrmat, annot=True, cmap='RdYlGn',  
square=True, linewidths=0.5)
```

```
plt.title('Correlation Matrix')
```

```
plt.show()
```